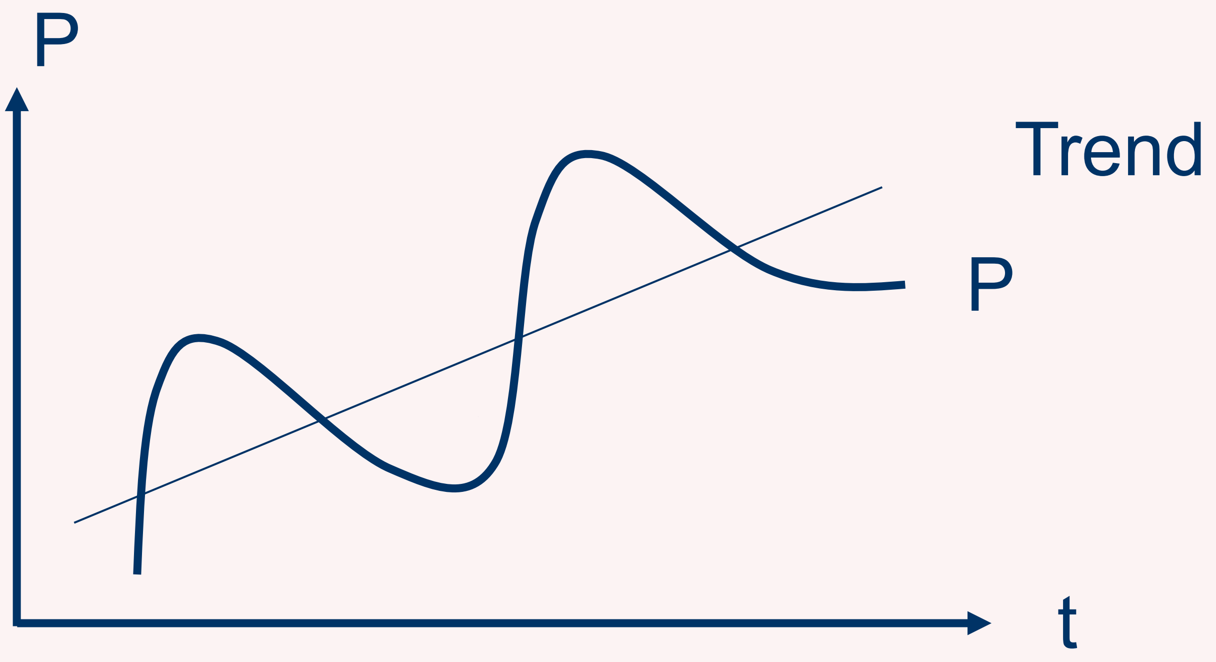
GY457
Applied Urban and Real Estate Economics

Block II—**Topic 6:**

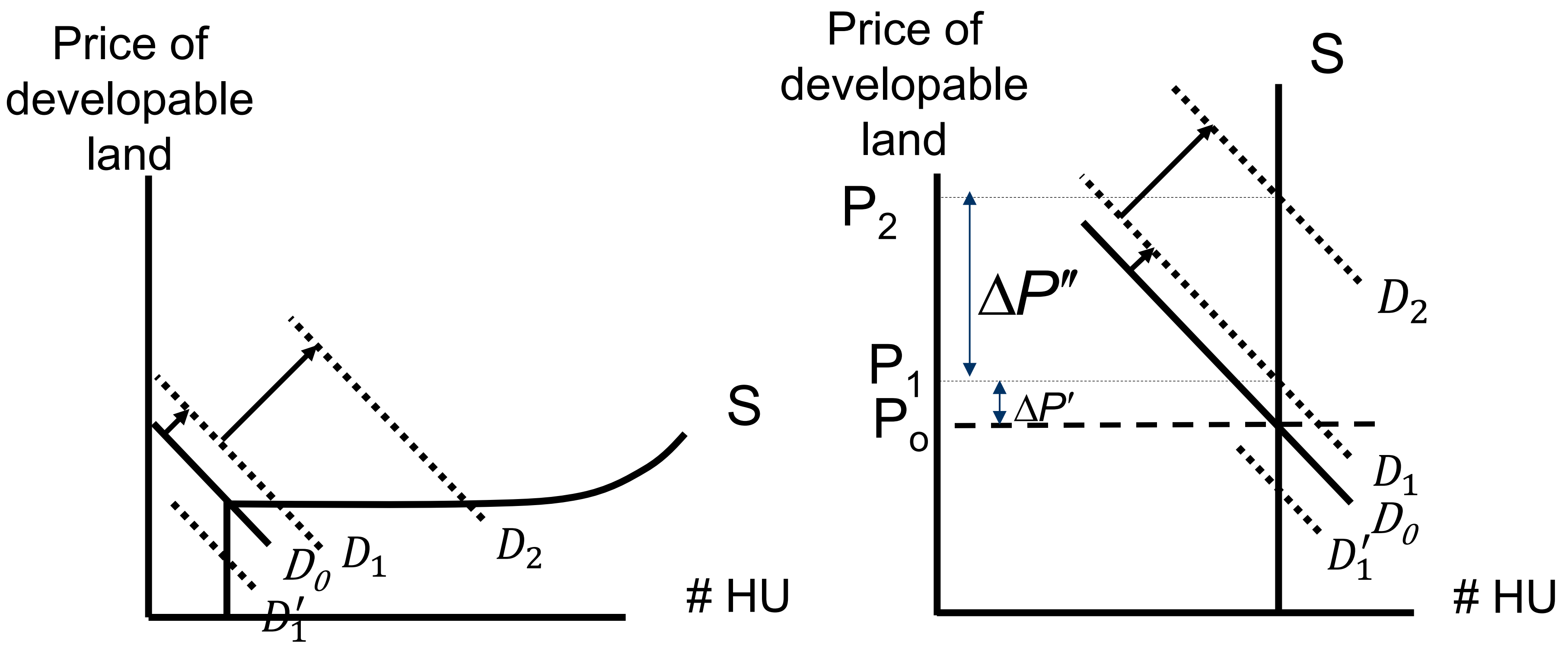
Endogenously Driven Real Estate Cycles and Behavioural Explanations



Last two lectures in a nutshell…

- Effect of demand volatility on land and house prices…

City with elastic supply ***City with inelastic supply***



This lecture

1. Some Further Stylized Facts and Puzzles
2. Endogenously Driven Cycles & Behavioural Explanations: Theories & Evidence
3. What About the Cyclicality of Rents and the Price-to-Rent Ratio?
4. Commercial vs. Residential Real Estate Cycles
5. Conclusions and Policy Implications

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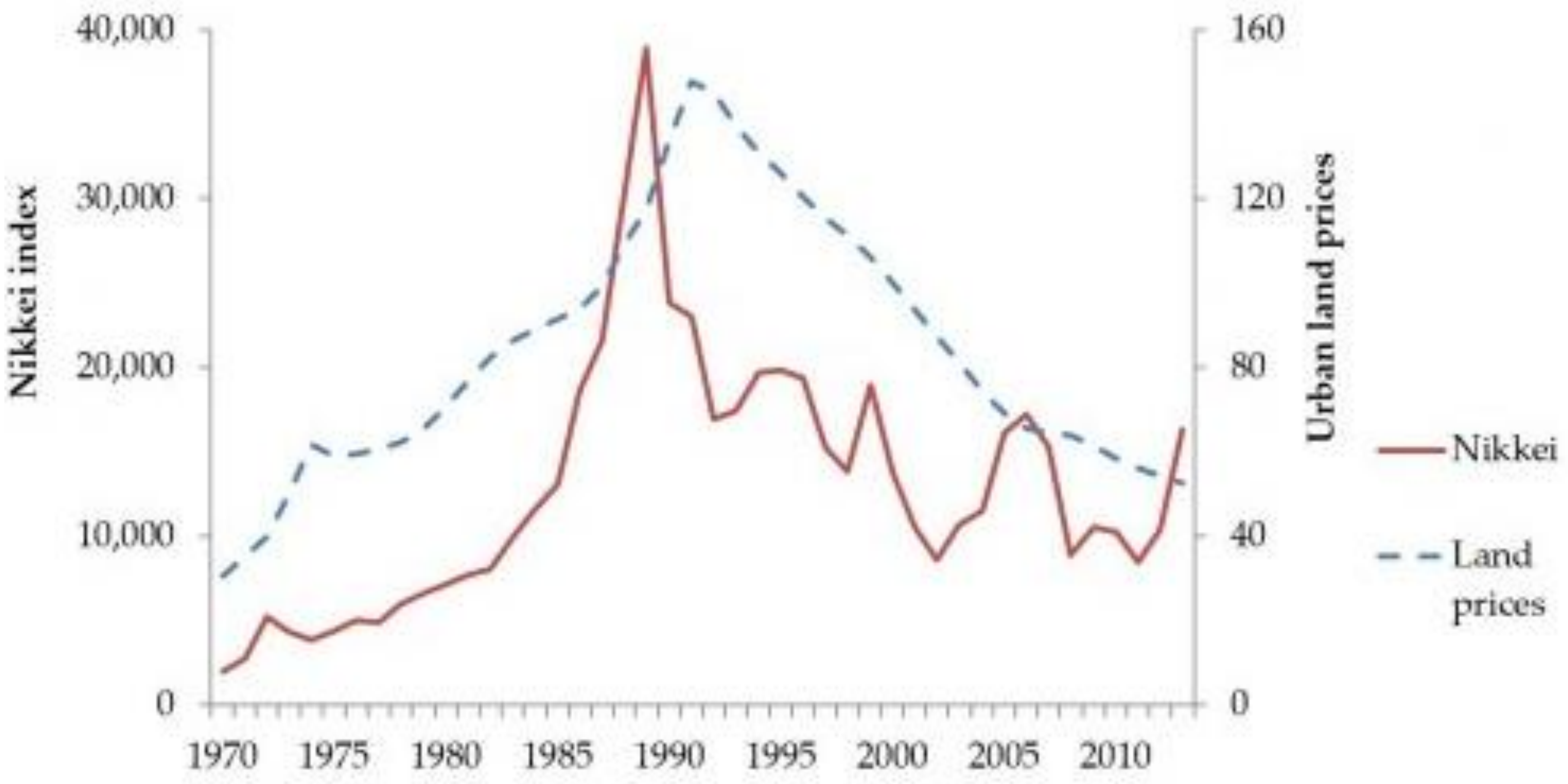
Puzzle: Japanese (property) asset bubble

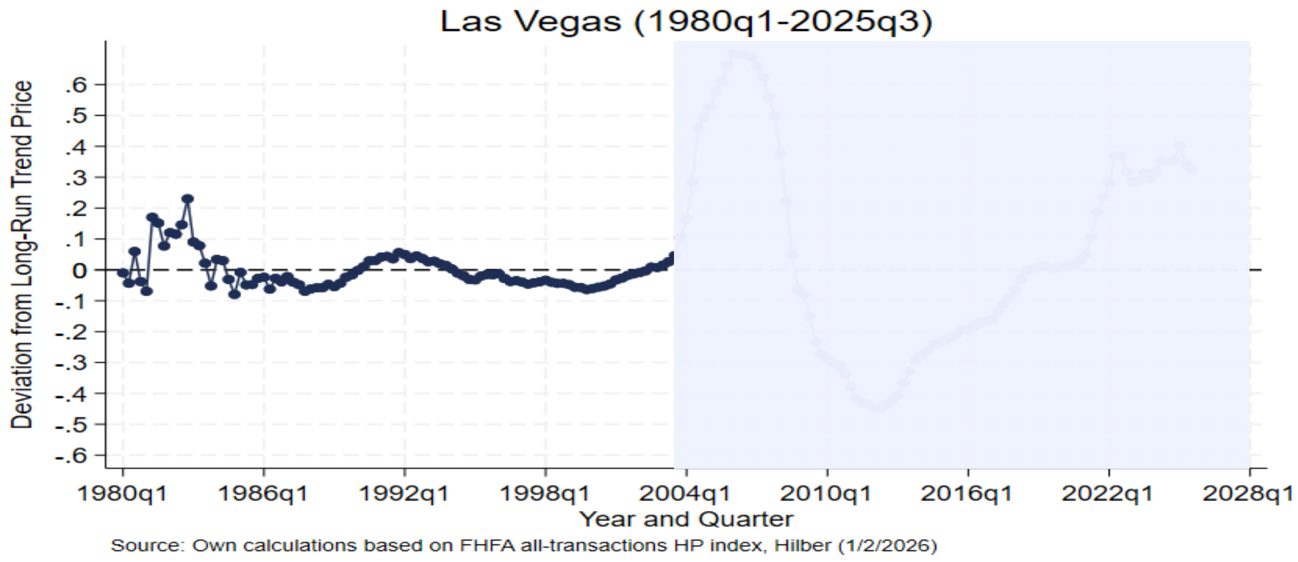
- **Urban land prices and stock prices 1970-2013**

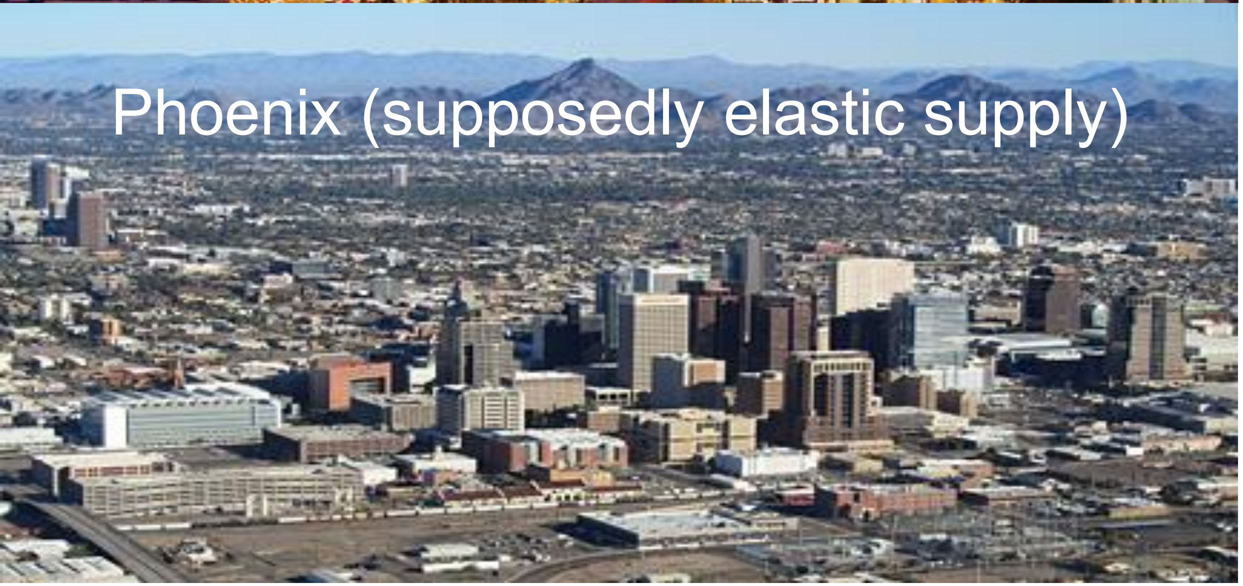


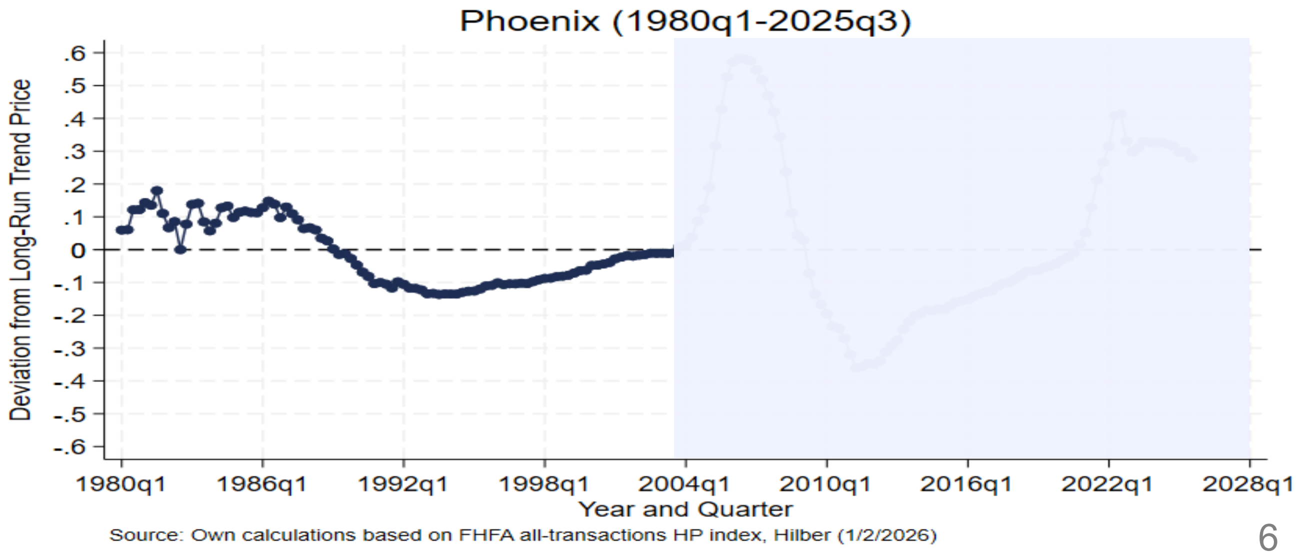
⇨ But population is only shrinking since 2011...

Sources: Nikkei and Japan Real Estate Institute (http://inflationmatters.com/japanese-deflation-myth/)

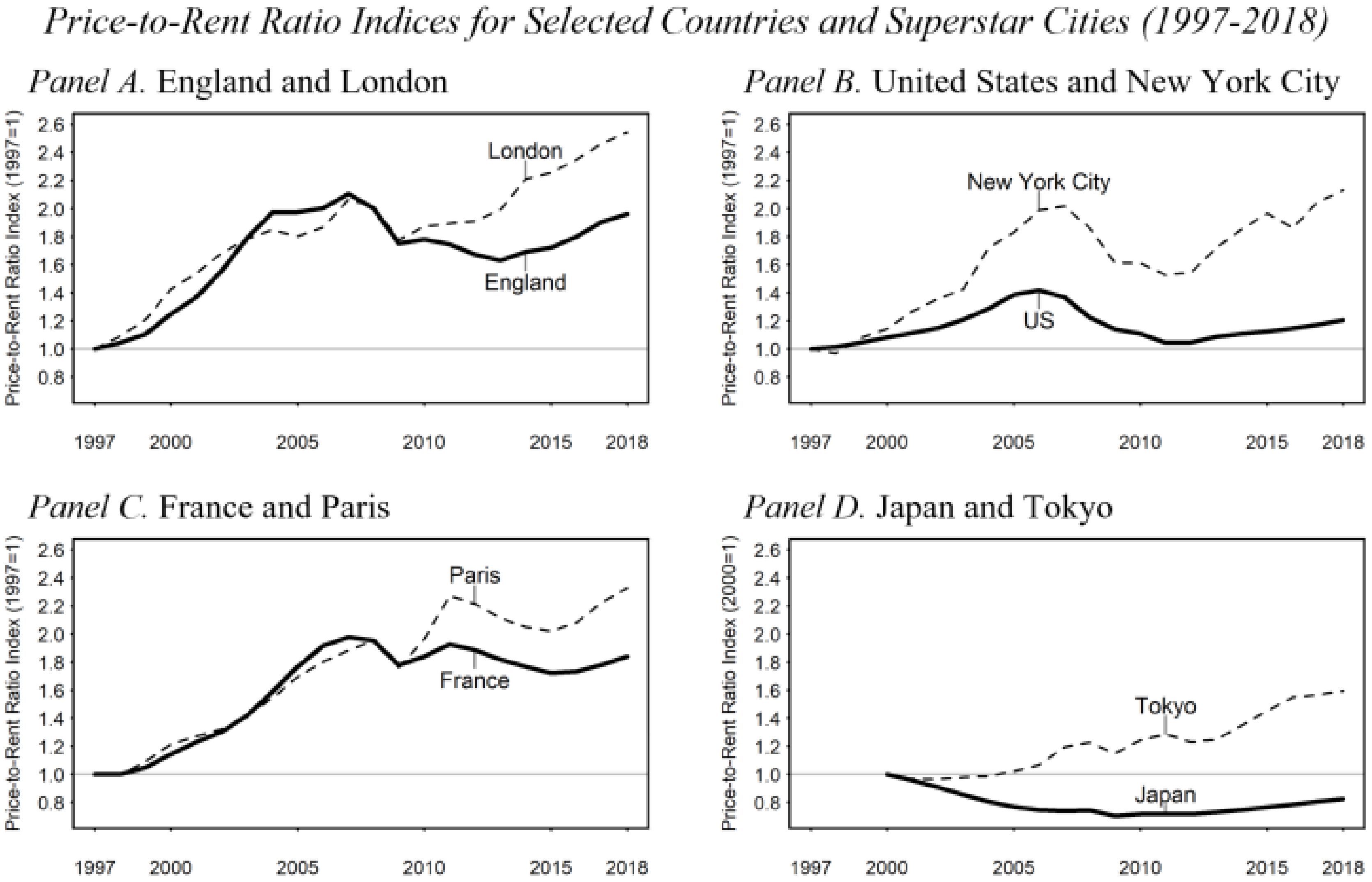
Some more puzzles...







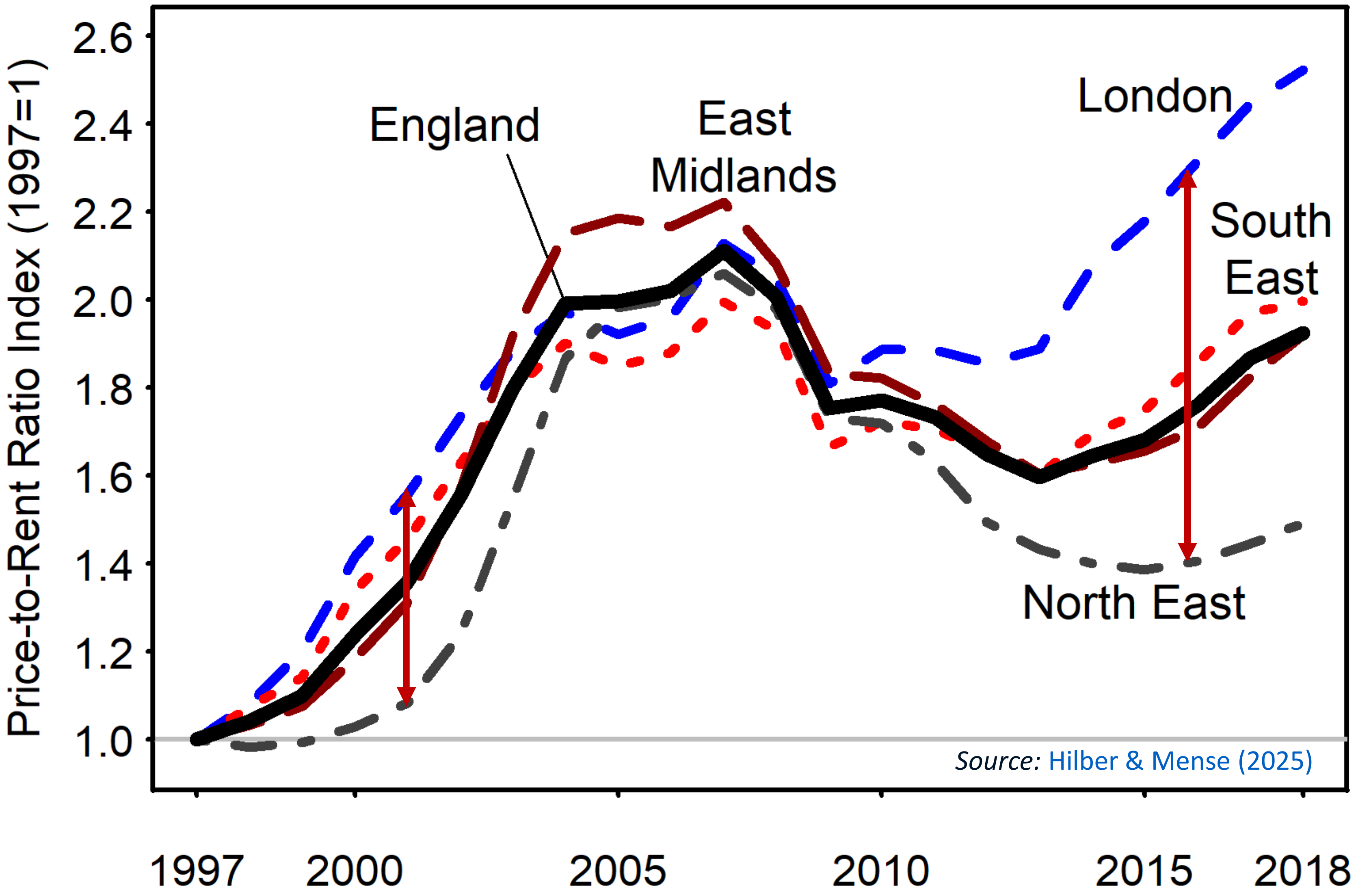
Another puzzle: Prices vs. rent dynamics



Japan hard to explain with interest rates!

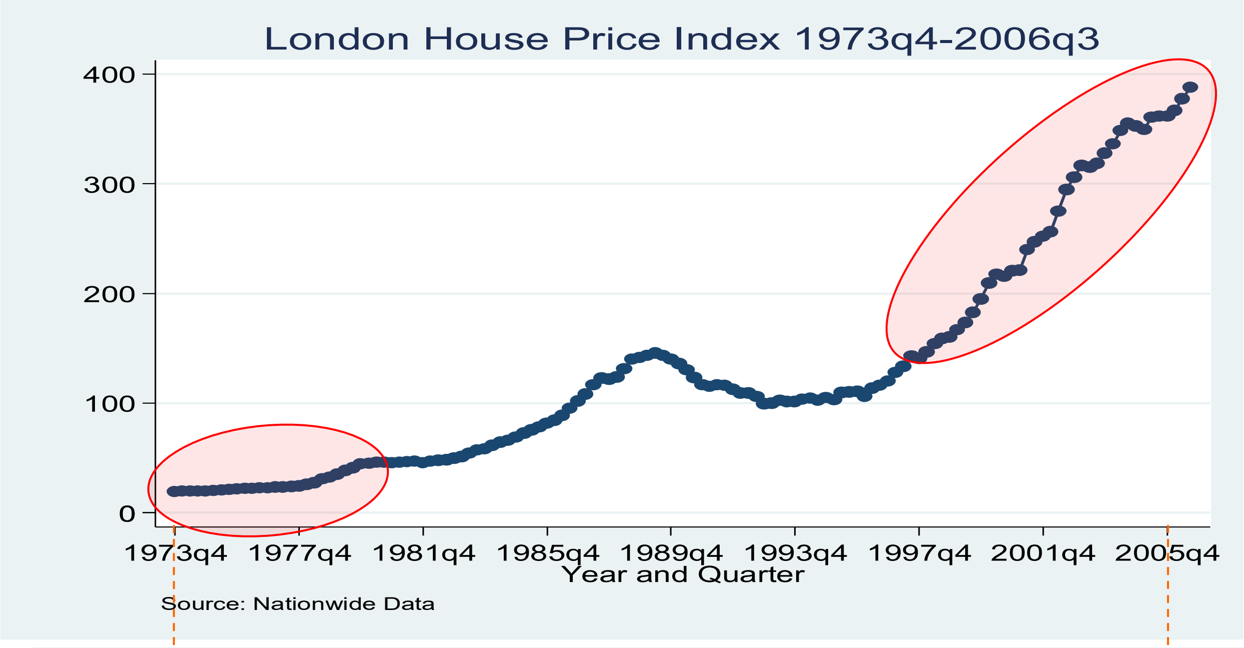
Source: Hilber and Mense (2026 EJ)

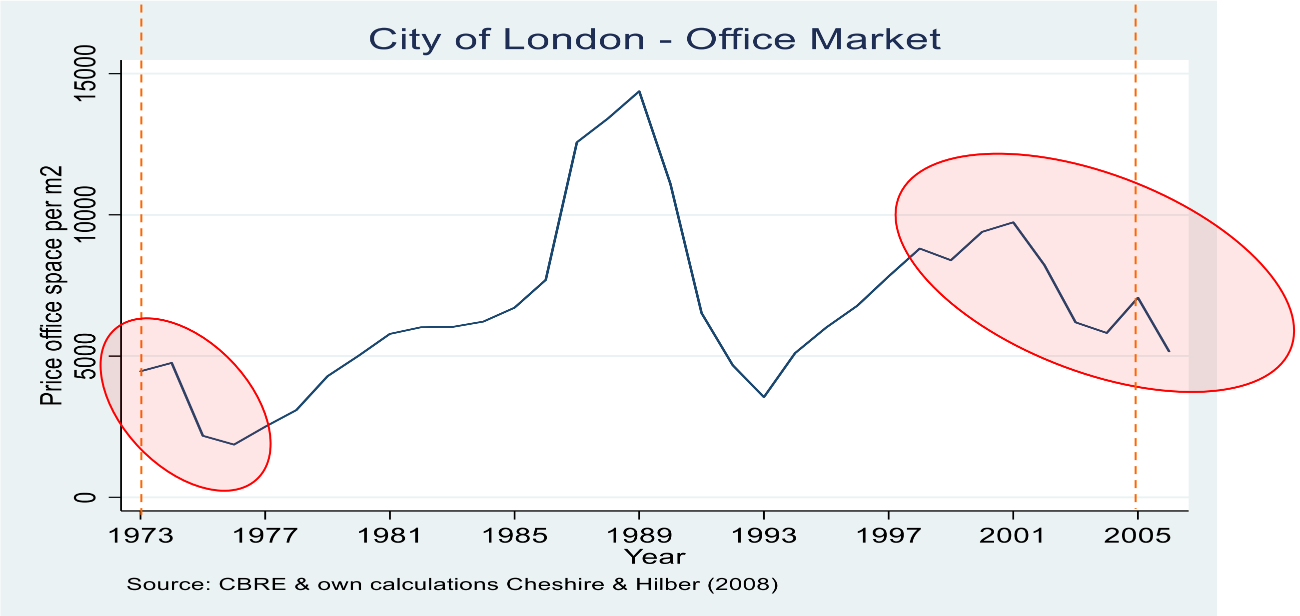
The spatial divergence in P/R-ratio is cyclical
(Small at peak & during downturns, but massive during upturns)



Difference hard to explain with interest rates!

A final puzzle: Residential vs. office market (in London)





How can we explain?
Revisit phenomenon of real estate cycles…

- What do **house prices** measure?
  - Forward-looking concept
  - If market participants have perfect foresight:

  Price = Sum of discounted future rents (and costs) associated with property/land [useful formula from finance: Gordon Growth Model: $P = \frac{R_1}{(i-g)}$]

- So far, assumed cycles are driven by **repeated exogenous demand shifts** (=business cycles)
  - Economic boom ➪ property price boom
  - Recession ➪ property price bust
  - Magnitude of **'exogenous cycles'** depends on supply price elasticity only

What did we ignore? (Discuss)

- **Lags** (planning, development, construction) ⇨ Often lagged adjustment
- Evidence of '**disequilibrium**': 'overbuilding' & high vacancy rates + cycles in transaction volume and 'time on market'
- **Mortgage markets** ⇨ downpayment & liquidity constraints
- Existence of **myopic agents, unrealistic expectations** & other **'behavioural aspects'**
- **Transaction costs** and other market imperfections (i.e., assumed efficient markets)

⇨ May give rise to **'endogenous cycles'** – initial shock may trigger endogenous oscillations (=cycles independent of exogenous shocks)

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Explore six alternative explanations one by one...

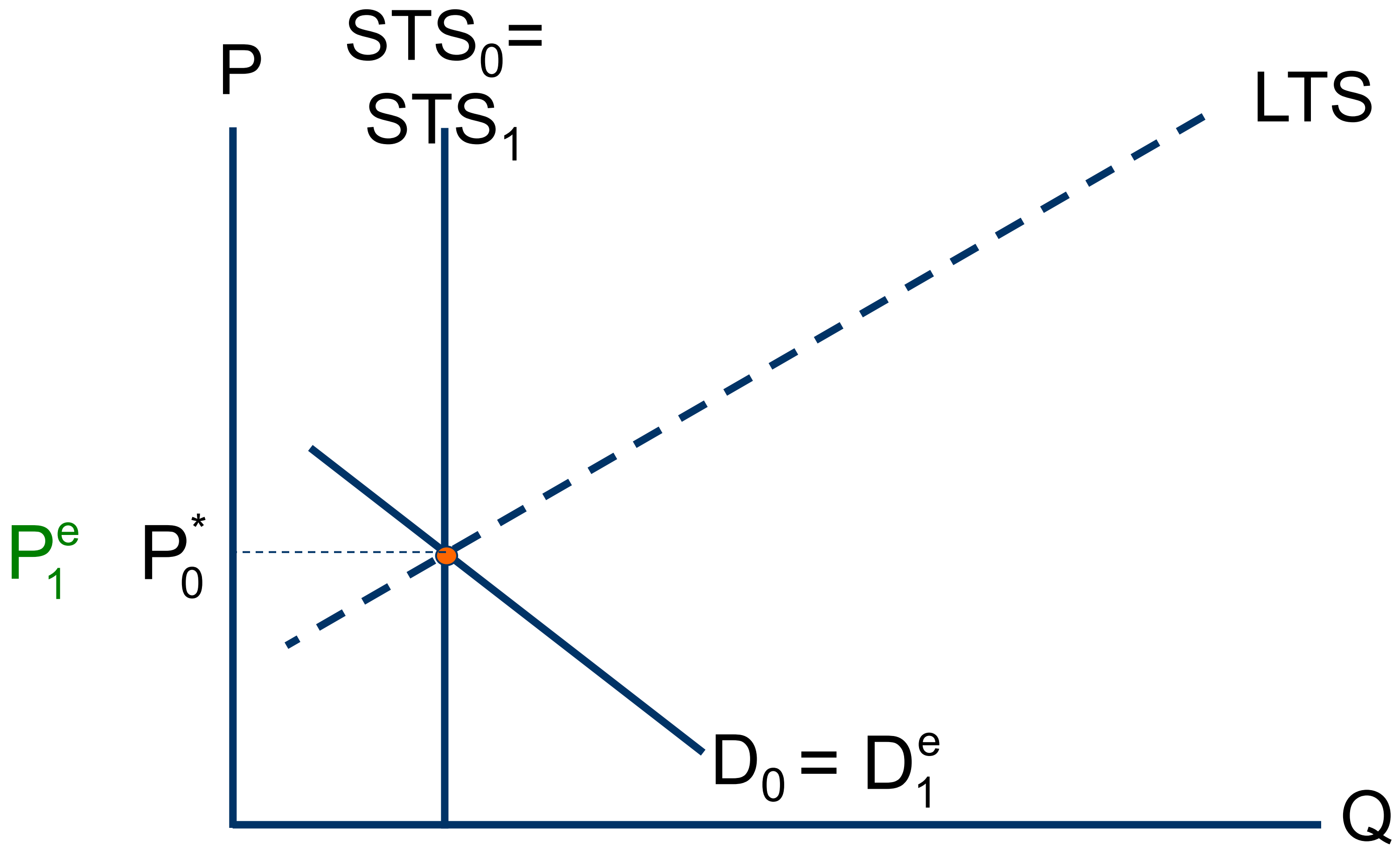
1. Myopic agents (developers & lenders) + lags (unrealistic expectations of agents on *supply side*)
2. Irrational exuberance (euphoria of investors) (unrealistic expectations of agents on *demand side*)
3. Liquidity constraints
4. Loss aversion
5. Option theory and investment lags
6. Search theory & matching

1. Myopic agents & lags (Hog cycle, stock flow models, cobweb)

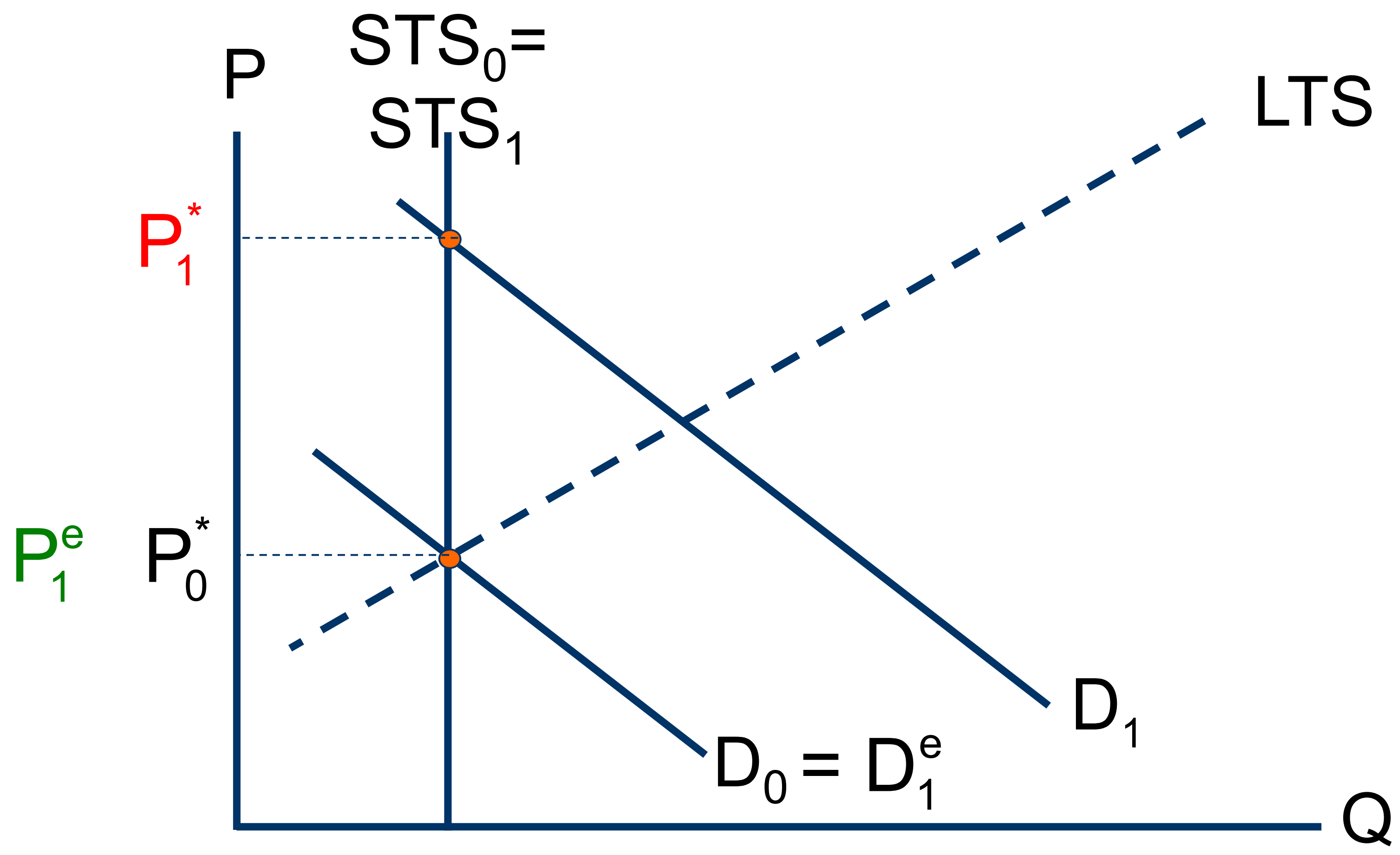
- Idea
  - Starting point: Unanticipated increase in demand *(e.g., driven by credit supply shock)*
  - Strong increase in prices due to **short-term supply shortage**
  - **Myopic developers and mortgage lenders** base decisions on observed prices

➔ **Consequence?**

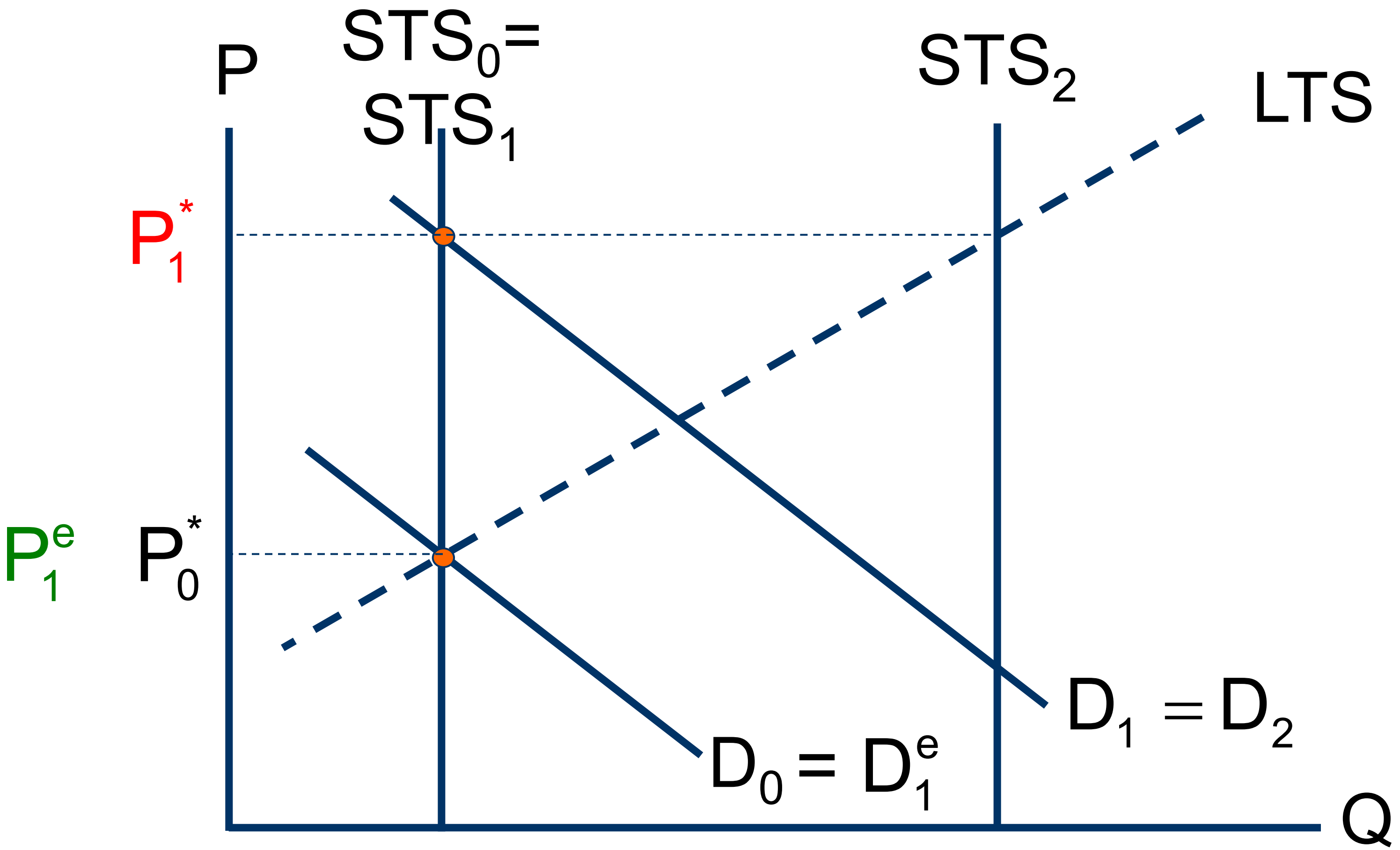
Myopic agents & lags (cont.)



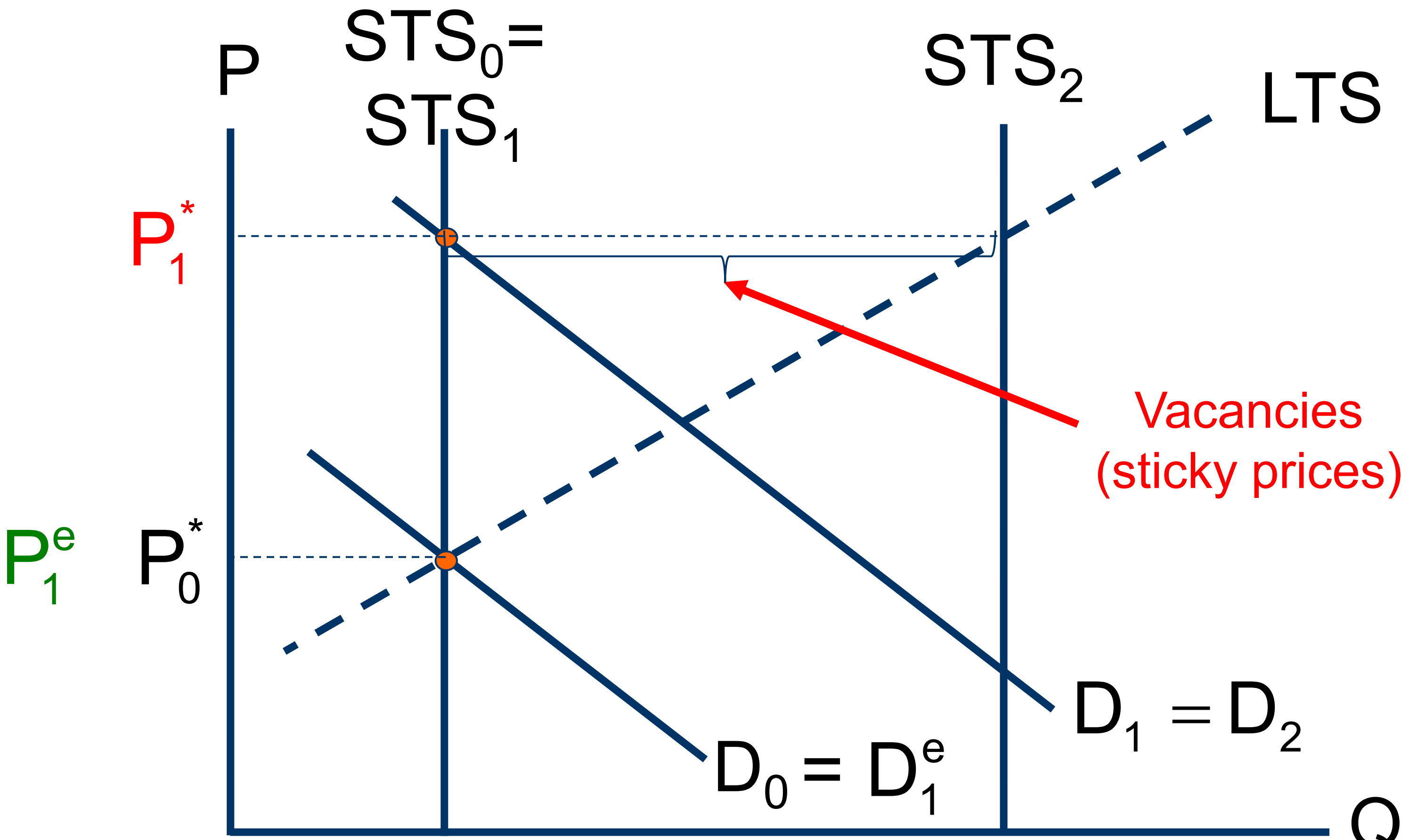
Myopic agents & lags *(cont.)*



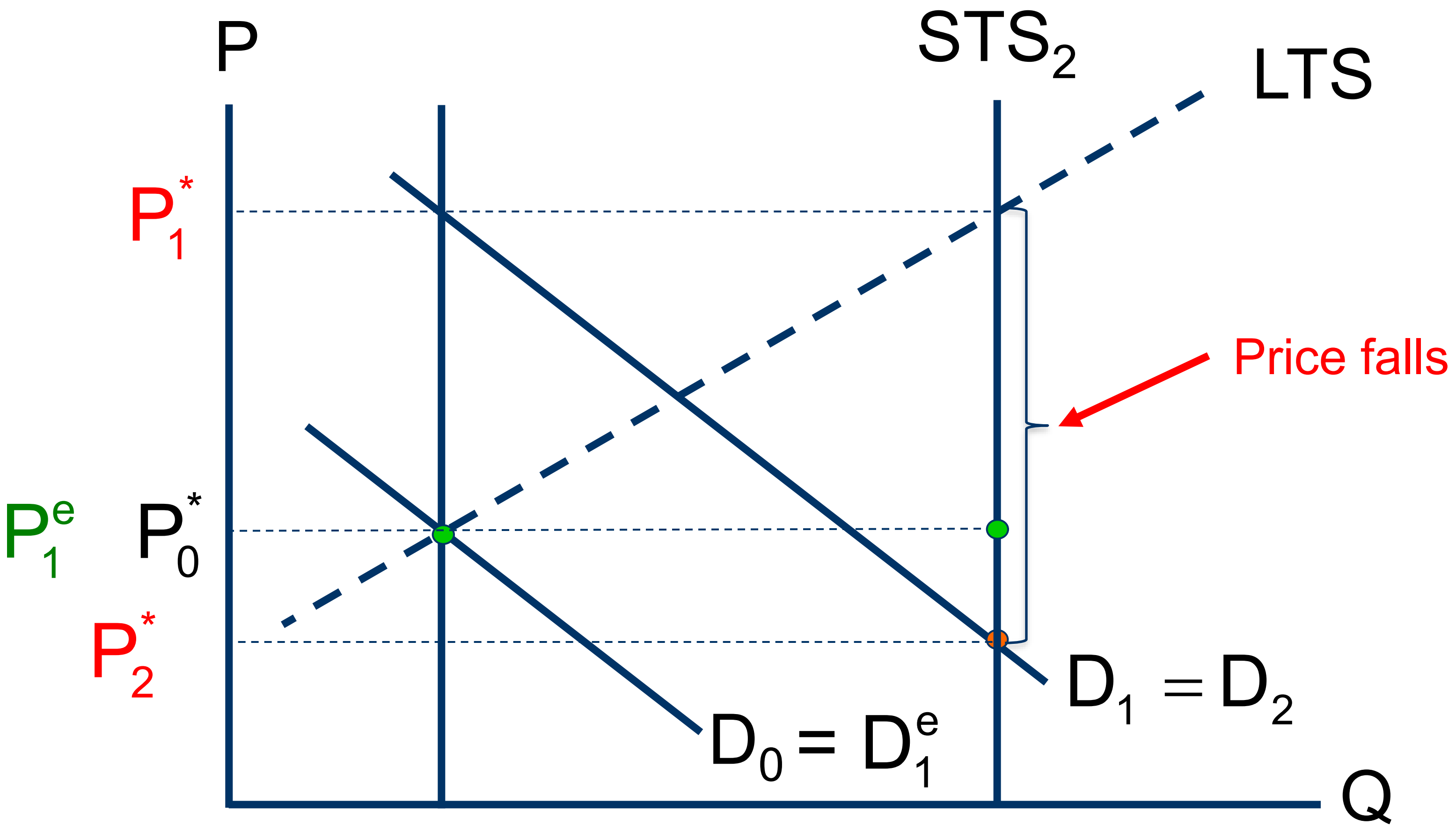
Myopic agents & lags (*cont.*)



Myopic agents & lags (cont.)



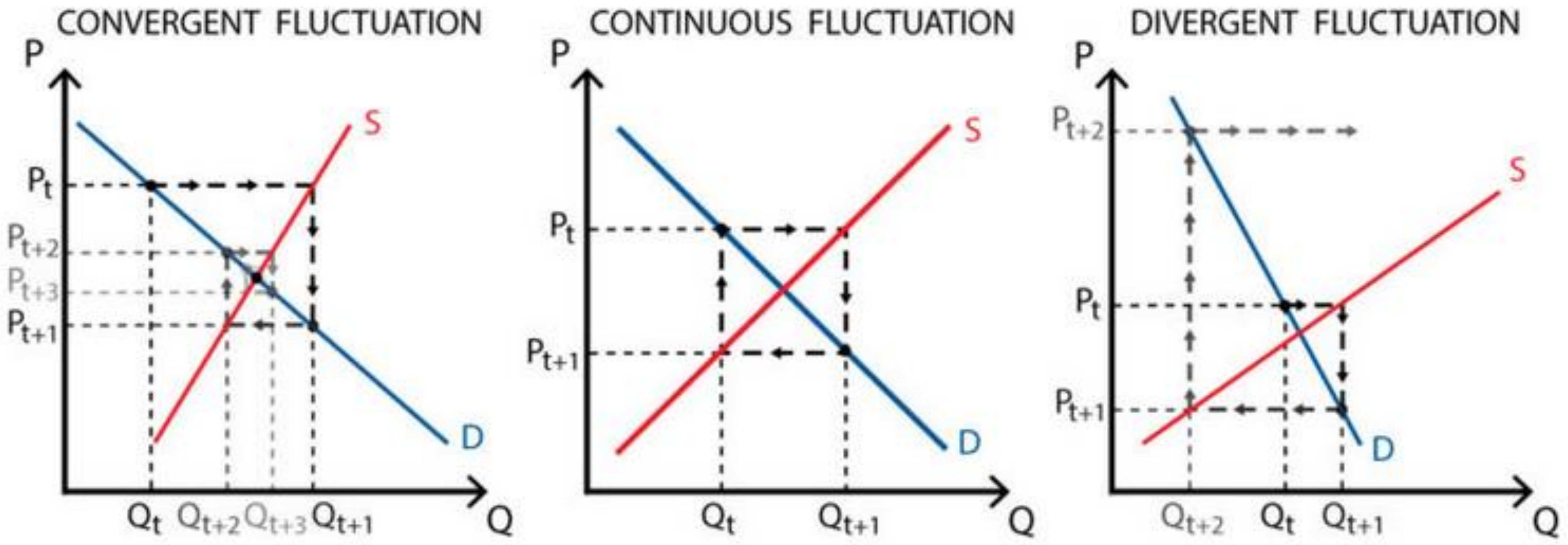
Myopic agents & lags (cont.)



Predictions

- In places with elastic long-term supply, unexpected strong positive demand shocks lead to
  - Significant overbuilding (high vacancy rates) if prices are sticky or
  - Drop in property prices or
  - Both
- In commercial RE: Excess supply greater if existing tenants have long-term leases (i.e., strongly lagged responses)
- Induced cyclicality more pronounced if long-run supply is very elastic relative to demand – Why?

Cobweb model
(Price adjustment when in disequilibrium)



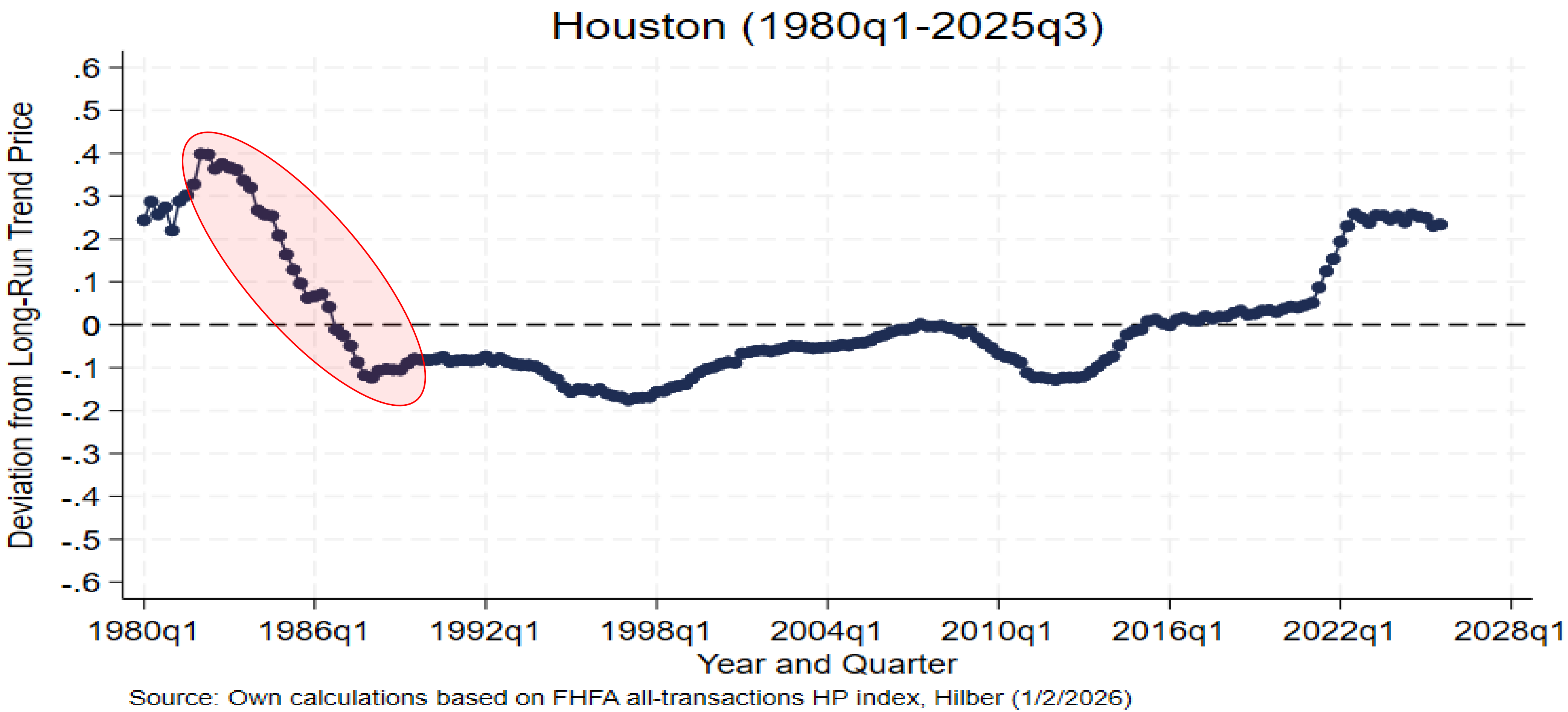
⇨ Shortcomings?

1. Long-run supply curve is arguably kinked
2. Are economic agents really permanently myopic?
   Can prices (permanently) diverge?

Empirical evidence

- Houston (TX) in 80s – anecdotic evidence
  - Unexpected boom in late 70s lead to **severe overbuilding** caused by myopic developers & mortgage lenders
  - Subsequent oil price shock and recession lead to **price collapse**

1980s Bust in Houston



⇨ **Why Houston?**

1. Has no zoning ordinances! Very elastic long-run supply!
2. Very dependent on oil ⇨ Very fluctuating demand

Critique

- **Relies on expectation errors** on the part of **supply actors** (developers, bankers)
- Even if supply actors are myopic, are they likely to constantly repeat mistakes?
- Not much rigorous evidence – evidence is mainly anecdotic

2. Irrational exuberance (euphoria)

- Idea
  - Investors observe strong **past** price increases
  - “Plausible story” tells them that price increases will go on forever
  - **Excessive/unrealistic public expectations** of future price increases start to form
  - Buyers **become euphoric** and increase their reservation prices

2. Irrational exuberance *(cont.)*

- **Herding behaviour of investors** (see last seminar!) further spurs demand
  - Raises prices (vicious circle) ultimately creating ‘bubble’
  - **‘Bubble’:** Refers to a situation in which **excessive** public expectations of future price increases cause prices to be temporarily elevated (Case and Shiller 2003)

Predictions

- Property prices can strongly deviate from values that are supported by fundamentals
- “Bubbles” ultimately end in price crash
- Overbuilding in markets with long-run elastic supply

Empirical evidence

- Tulip Bubble in 17th Century
- Internet equity-bubble in late 1990s
- **In real estate?**
  - Asset bubble in **Japan** (incl. RE?) **during late 1980s**
  - **Indirect evidence** from survey results (Case & Shiller 1988, 2003, Case, Shiller & Thompson 2012) and "information experiments" within panel-surveys (Armona *et al.* 2018)
  - Capozza *et al.* (2004): Show that **serial correlation is stronger in booming markets** consistent with 'euphoria' and backward-looking expectations

Critique

- **Non-falsifiable:** Theory is long on predictions but short on testable hypotheses
  - Can look at residuals (actual price minus 'fundamental price') ➪ But is this evidence for euphoria or for OVs or model misspecification?
  - Survey evidence only very indirect
- Theoretical arguments
  - Are purchases and sales in housing markets really mainly driven by investment (rather than consumption) motives? And are homebuyers really 'euphoric'?
  - High transaction costs should reduce speculative incentives

3. Liquidity constraints

- Developed by Stein (1995) & Ortalo-Magne & Rady (2005)
- Idea
  - Income shock strongly affects ability of potential first-time buyer to qualify for mortgage on a starter home
  - If income ↑ ⇨ demand ↑↑ ⇨ HP↑↑ ⇨ capital gain for existing owners ↑↑ ⇨ demand for trade-up home ↑↑ …
  - Can have dramatic impact on overall housing market

Empirical evidence

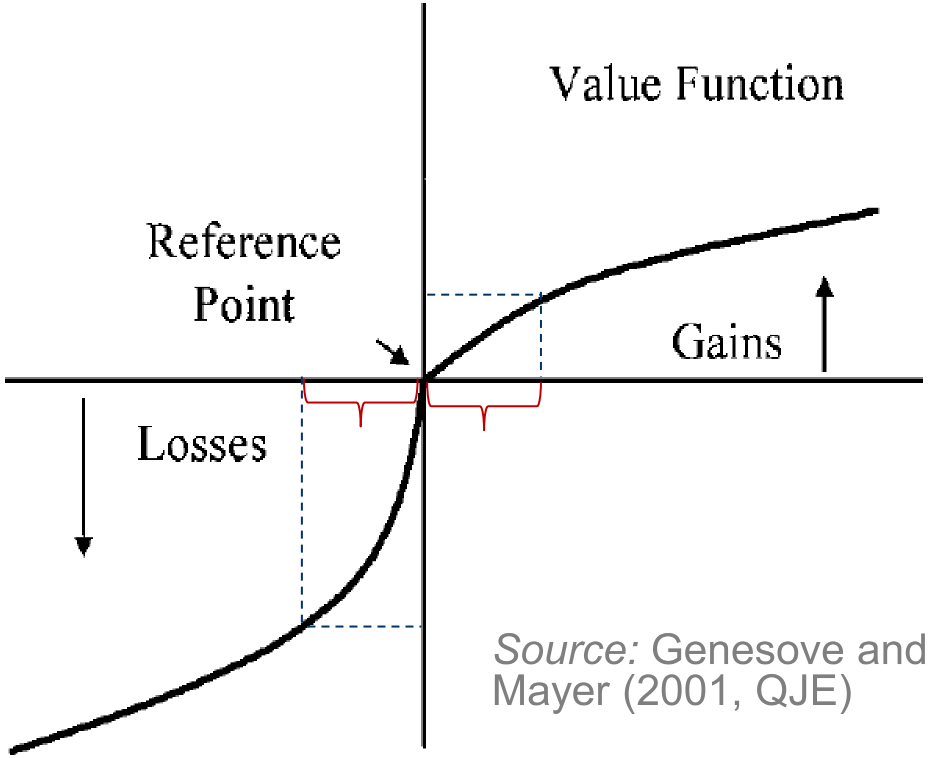
- Lamont & Stein (1999)
  - In cities with a large fraction of highly leveraged homeowners (first-time buyers), HP react more sensitively to city-specific shocks
- Genesove and Mayer (1997 AER)
  - High LTV homeowners set higher asking prices (because need to be able to buy next home)
  - Have longer expected time on market &
  - Ultimately sell at higher price

Critique

- **Only applies to residential RE**
  - Cannot explain commercial RE cycles, yet they are even more pronounced
- **Assumes no role for developers—cannot explain overbuilding phenomenon**
- **Alternative explanations**
  - ‘Leveraged cities’ might also be places with more inelastic supply of housing (untested)
  - Findings might be due to ‘loss aversion’….

4. Loss aversion

- Theory developed by Kahneman & Tversky (1991)
- Applied to real estate by Genesove & Mayer (2001)
- Idea
  - Property owners are loss averse and are not willing to sell with loss in downturn



Source: Genesove and Mayer (2001, QJE)

Predictions

- Sellers' reservation prices are less flexible downward than buyers' offers
  - Seller characteristics (loss aversion) affects transaction prices
  - Transaction volume falls and time on market increases when prices decline

Empirical evidence

- Genesove & Mayer (2001, QJE)
  - Loss aversion matters a lot
  - Liquidity constraints still matter, but much less than previously thought
  - Listing price only affected if seller is severely downpayment-constrained (LTV>0.8)

Critique

- Similar to liquidity constraints
  - Can also not explain overbuilding phenomenon: no role for developers
  - Can it explain commercial cycles? Are (profit maximizing) developers and top managers of firms loss-averse?
  - Can only explain dynamics of downturn but not full cycles

5. Option theory and investment lag

- Theory developed by Grenadier (1995)
- Key idea
  - Consider profit maximising owner of land
  - What is the optimal timing to exercise the option to develop and the option to rent out extra units?

Prediction

- Increase in demand volatility
  - Increases value of **option to wait** renting out
  - Makes **excess capacity** more profitable

Empirical evidence

- Real estate markets with most volatile demand (office) display greatest degree of vacancy rate stickiness (because option to wait renting out is more valuable)
- Existence of building booms (excess capacity) in face of declining demand and property values

Critique

- Does good job explaining sticky vacancy rate and overbuilding phenomena but less good at explaining other phenomena
- Evidence is largely consistent with theory but not absolutely conclusive

6. Search theory and matching

- Builds on work by Mortensen, Diamond & Pissarides, first applied to real estate by Wheaton (1990), refined by Head *et al.* (2014)
- Idea
  - Housing is heterogeneous good and gathering information on properties is costly
  - Buyers expend costly search effort to find better house, while sellers either hold two units (old and new) until buyer is found or, alternatively, they temporarily rent while selling old house

Mechanism (as proposed by Head *et al.* 2014)

1. **Income shock** spurs immediate increase in house search as HHs (buyers) enter
2. It takes time for buyers to find suitable houses and for construction to respond
3. To meet immediate housing demand, **vacant houses are shifted to rental market** ➪ tightness of owner-occupied market rises ➪ Sales price ↑
4. Eventually: Construction ↑ ➪ vacant homes ↑
5. As income reverts to long-run level, stock of buyers declines ➪ buyer-to-seller ratio falls ➪ price reverts to steady-state

Evidence

- Diaz & Jerez (2013) & Head *et al.* (2014)
  - Income shocks cause prices, construction levels and vacancy rates to respond cyclically, consistent with search & matching mechanism

- Ngai & Tenreyro (2014)
  - Seasonal moving patterns and weather fluctuations cause “hot” and “cold” seasons in housing market, consistent with search & matching

Critique

- Mainly applies to residential RE
  - Cannot really explain commercial RE cycles, yet they are even more pronounced
  - Some price cycles are not associated with strong cyclicality in vacancy rates or construction

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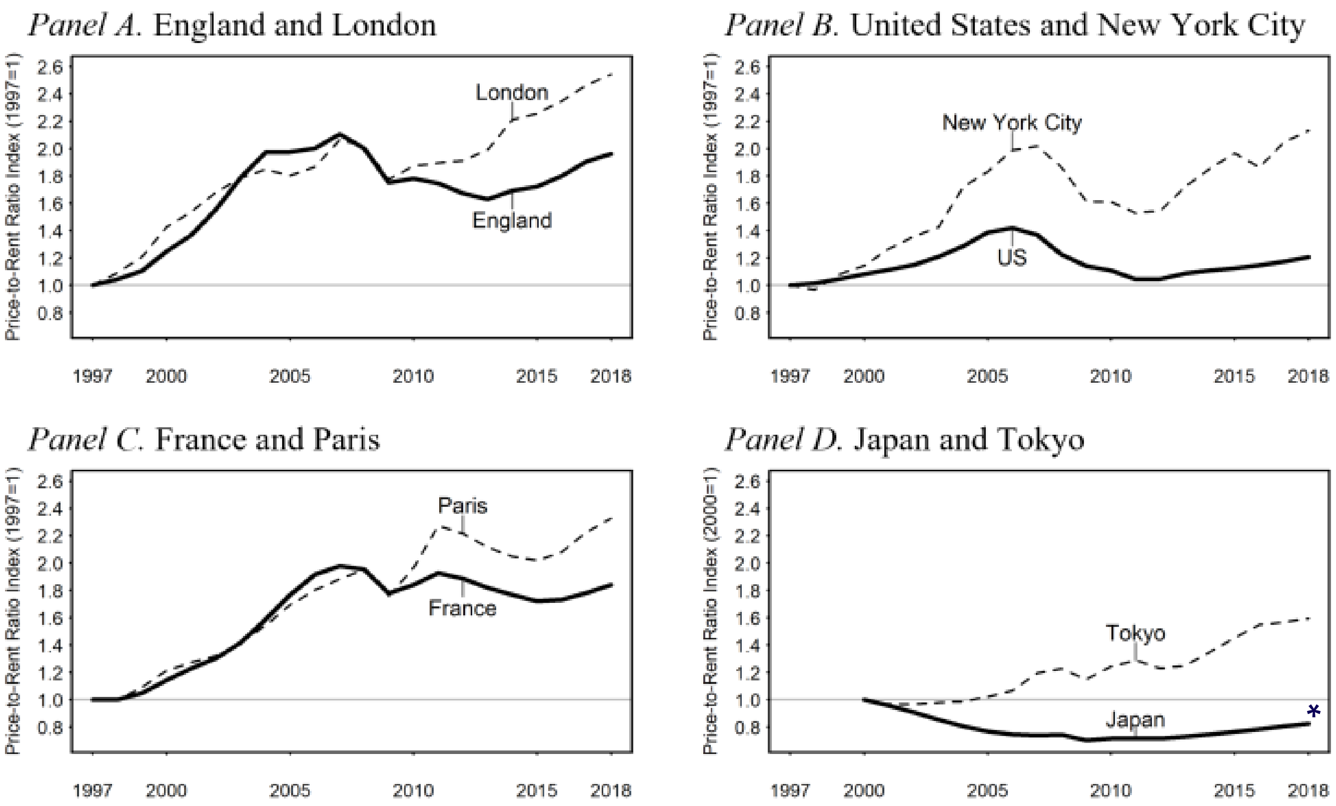
Discuss...

- Why have house prices risen so much more than rents over the last 20 years and why particularly in superstar cities?

Figure 1

Price-to-Rent Ratio Indices for Selected Countries and Superstar Cities (1997-2018)

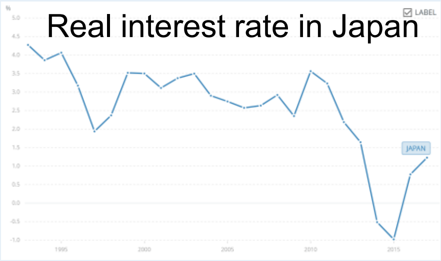


** Despite falling real interest rates b/w 2000-18*

Source: Hilber and Mense (2024)

Possible explanations? (Discuss)

- **Macro factors: Falling interest rates**
  - But why particularly in superstar cities? Why decrease in P/R-ratio in Japan? Why cyclical?
- **Changing credit-conditions**
  - But credit supply deregulation *in UK* prior to 1997
  - Why particularly in superstar cities?



Source: Worldbank

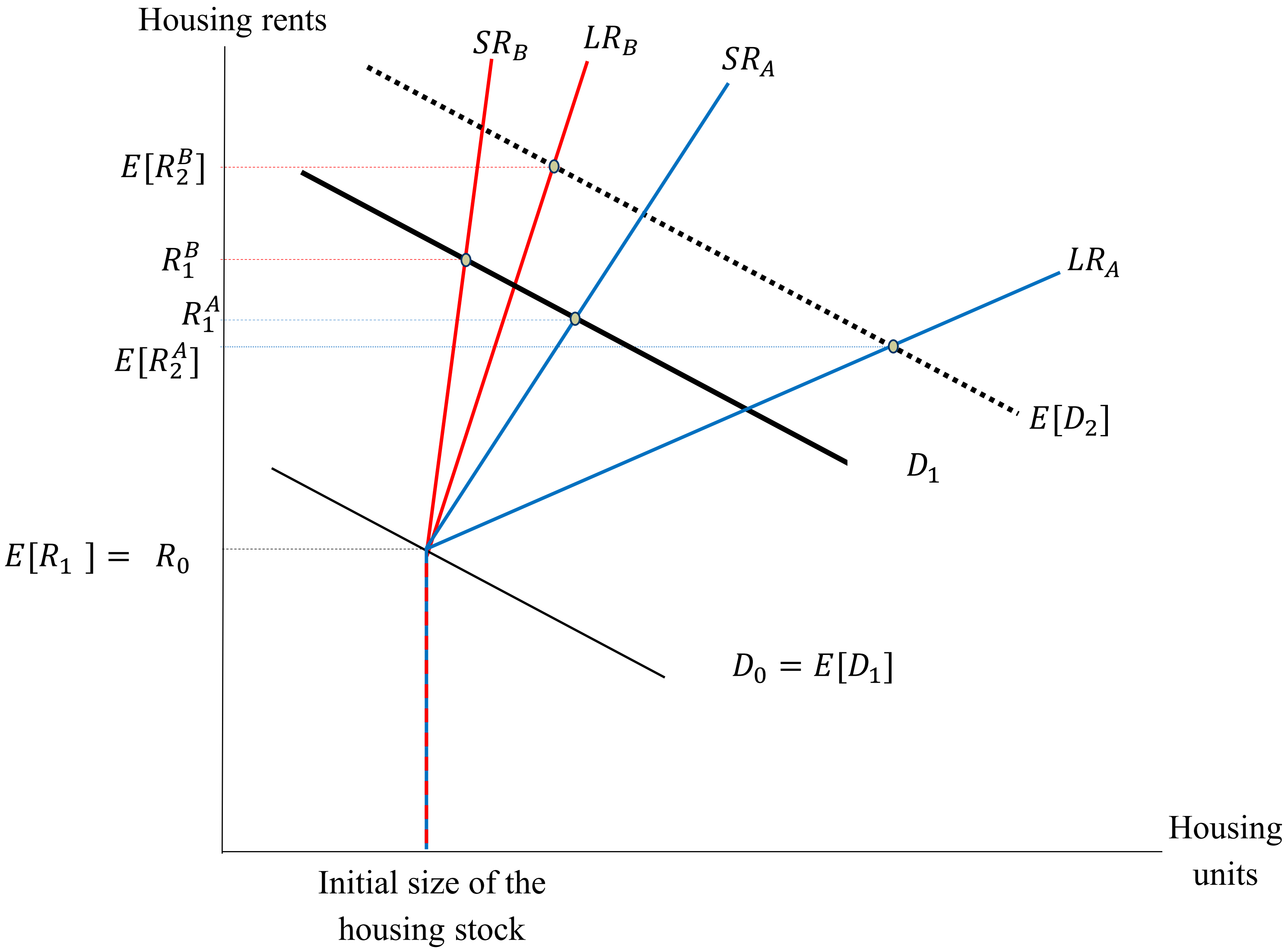
- ***Unrealistic* expectations? (Irrational exuberance)**
  - But why does P/R-ratio not collapse once ‘bubble’ bursts (here rising over extended period)
- **Rising inequality – rich own, poor rent (separate markets)**
  - But why cyclical?
- **Global investor demand (for superstar cities)?**
  - But seems relevant only in desirable neighbourhoods in superstar cities + tourist areas
- **Measurement issue:** HPs = transaction prices, rents from surveys ⇨ landlords often only increase rents significantly when tenant moves
  - But can this really explain massive and cyclical increase in P/R-ratio over 20 years? (Market rents should fully adjust eventually)

An alternative explanation… (Based on Hilber & Mense 2026 EJ)

- Consider a setting with short- and long-run supply constraints
  - For any given location: Supply is more inelastic in short-run
  - Some locations (A) have more elastic (SR+LR) supply than others (B)
  - To fix ideas: A is Newcastle and B is London
- Assume housing demand shocks exhibit serial correlation… (which is a feature of the data!)
  - Consider model where housing has durability of two periods

An alternative explanation….
Consider positive demand shock

Period 1: SR; Period 2: LR



Starting point: Long-run equilibrium (no demand shock):

$$E[R_1] = R_0$$

$$P_0 = R_0 + r\,E[R_1]$$

$$\frac{P_0}{R_0} = \left(1 + r\frac{E[R_1]}{R_0}\right) = (1+r)$$

Positive demand shock:

Elastic case A:

$$P_1^A = R_1^A + r\,E[R_2^A]$$

$$\frac{P_1^A}{R_1^A} = \left(1 + r\underbrace{\frac{E[R_2^A]}{R_1^A}}_{<1}\right) < (1+r)$$

Inelastic case B:

$$\frac{P_1^B}{R_1^B} = \left(1 + r\underbrace{\frac{E[R_2^B]}{R_1^B}}_{>1}\right) > (1+r)$$

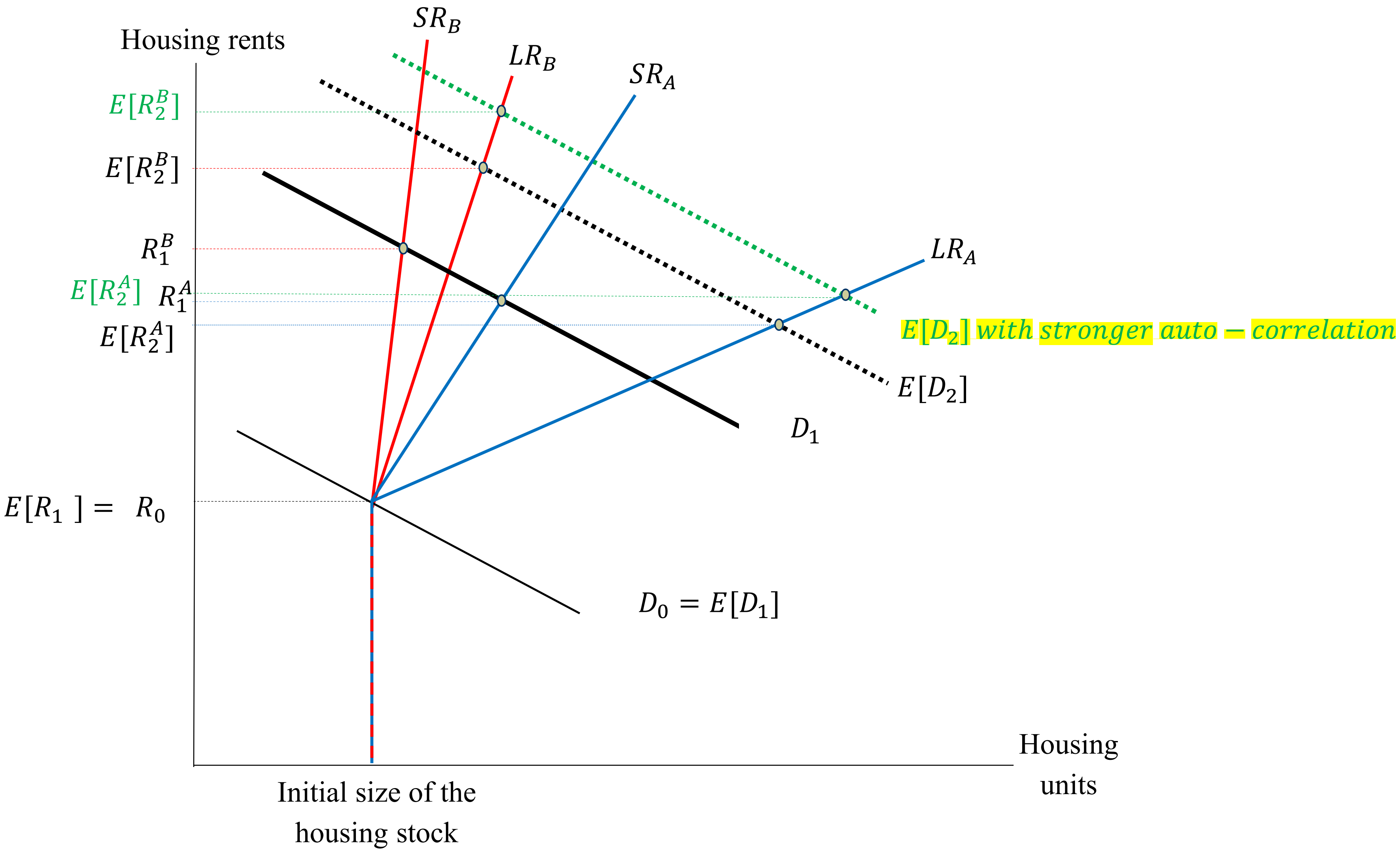
Predictions

- **Prediction 1:** If (long-run) supply is sufficiently inelastic and demand serially-correlated, the price-to-rent ratio increases in response to an initial positive demand shock.

- **Prediction 2:** This effect is more pronounced when local (long-run) supply is more price inelastic (⇨ superstar cities).

- **Prediction 3:** The interaction effect of demand shocks and the supply elasticity is amplified in phases with stronger auto-correlation of demand shocks (i.e., in the middle of local economic booms) (if persistence ↑, steeper supply curve leads to greater increase in $E[R_2]$)

Prediction 3 illustrated (role of auto-correlation)



Impact of Bartik-demand shocks on prices and rents
(Based on Hilber and Mense 2026 EJ)

	ΔLog House prices (IV)	**ΔLog Rents (IV)**
ΔLog (Labor demand shock)	0.28***	0.10
	(0.09)	(0.07)
ΔLog (LDS) × supply price elasticity	-0.76***	-0.095**
	(0.21)	(0.047)
ΔHtB dummy	0.017**	0.007***
	(0.007)	(0.002)
Year FEs	Yes	Year
Obs. (Number of LPAs)	7,211 (344)	7211 (344)
Sanderson-Windmeijer F, ΔLDS×elasticity	32.8	32.8

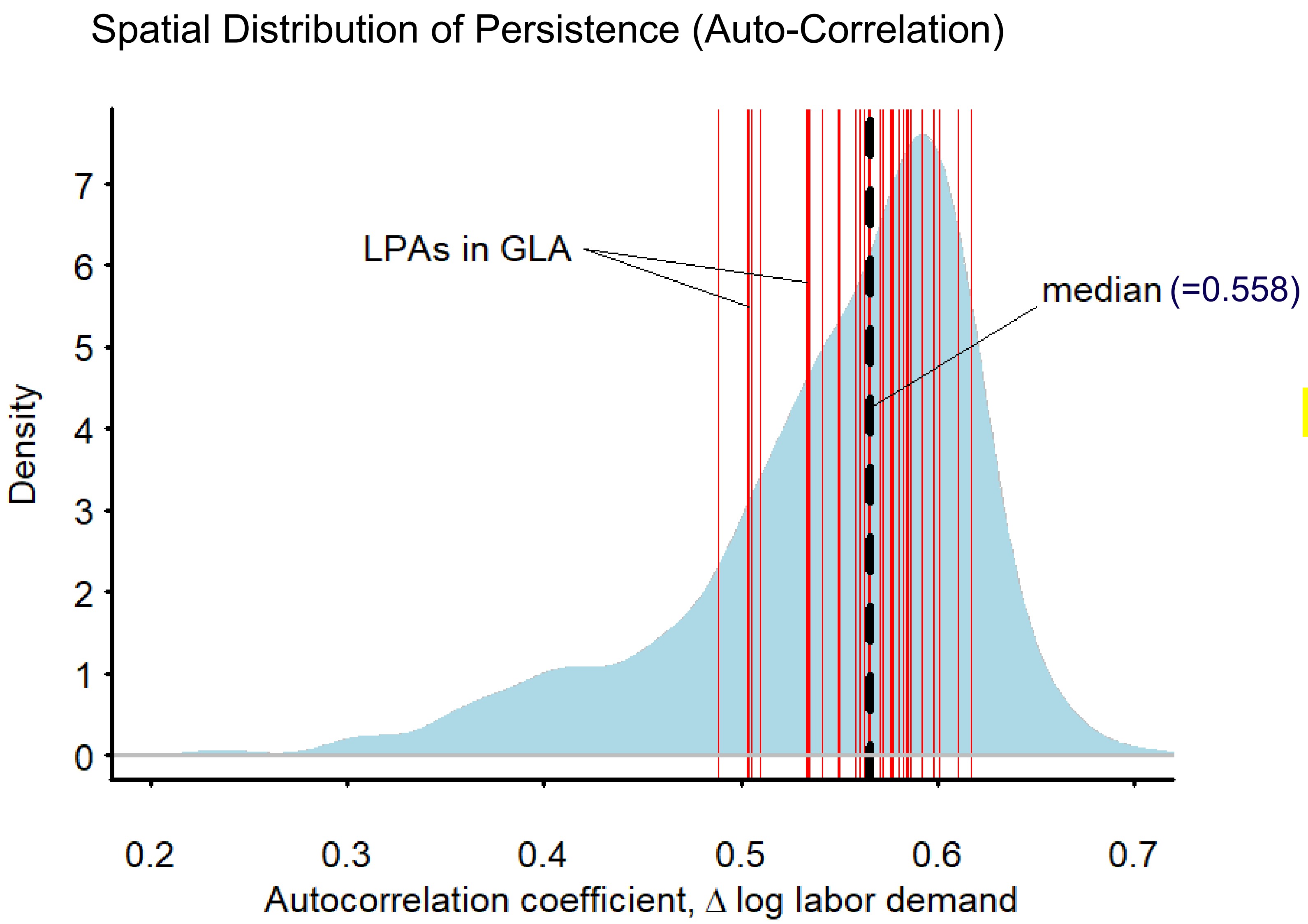
Notes: **Bold** var. are instrumented. Std. err. clustered using 2001 LPAs; ***: p < .01, **: p < .05.

Testing Predictions 1 & 2: Baseline specifications (2nd stage)

	ΔPrice-to-rent (IV)	
ΔLog (Labor demand shock)	22.81***	⇨ Prediction 1 ✓
	(10.8)	
ΔLog (LDS) × supply price elasticity	-60.2***	⇨ Prediction 2 ✓
	(9.0)	
Lags of ΔLog (LDS) & lags of interaction	Yes	
Δ Help to Buy dummy	Yes	
Year FEs	Yes	
Obs. (Number of LPAs)	7,211 (344)	
Sanderson-Windmeijer F, ΔLDS×elasticity	42.2	

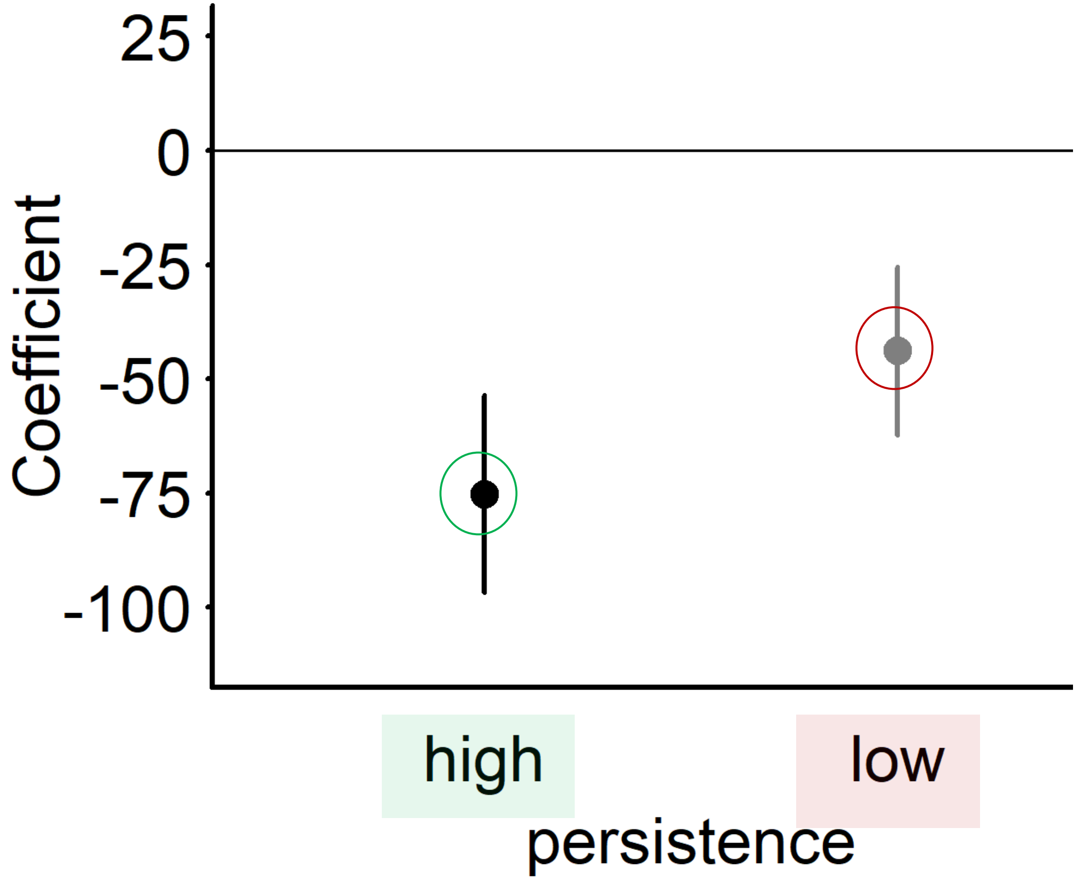
Notes: **Bold** var. are instrumented. Std. err. clustered using 2001 LPAs; ***: $p < .01$.

Testing Prediction 3: Persistence in local labor demand shocks



⇨Divide sample into above- and below-median persistence

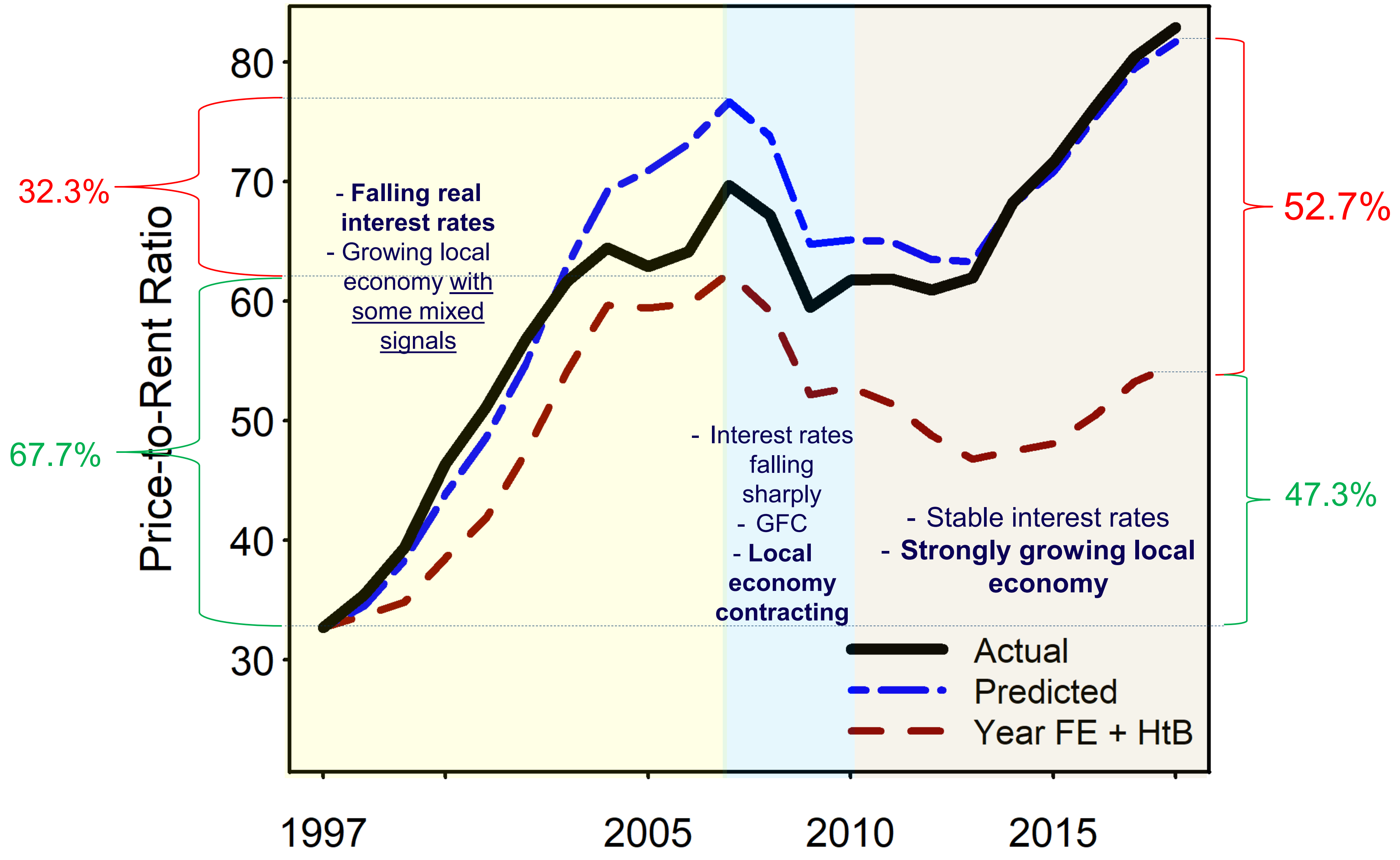
Testing Prediction 3: Regression coefficients of interaction effects for high vs. low persistence LPAs

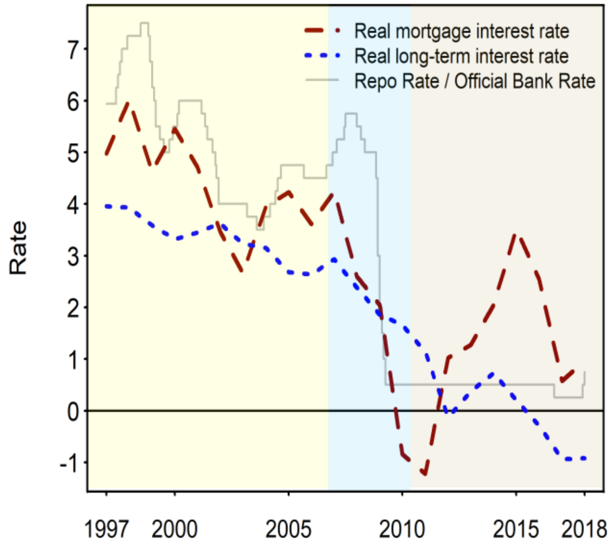


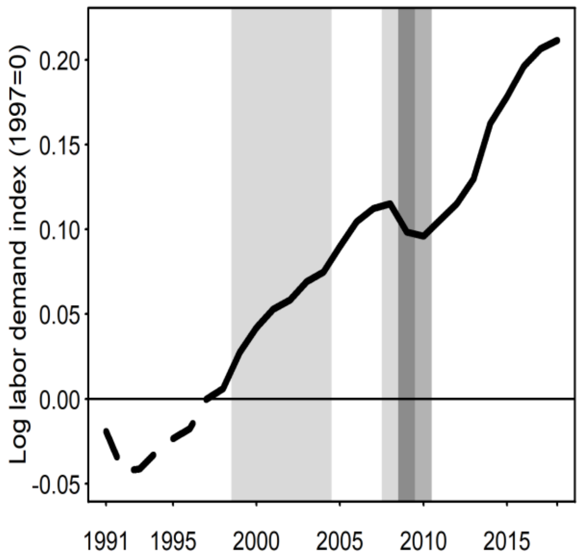
⇨ Prediction 3 ✓

Quantitative effects for London: Decomposition

(Based on empirical analysis)



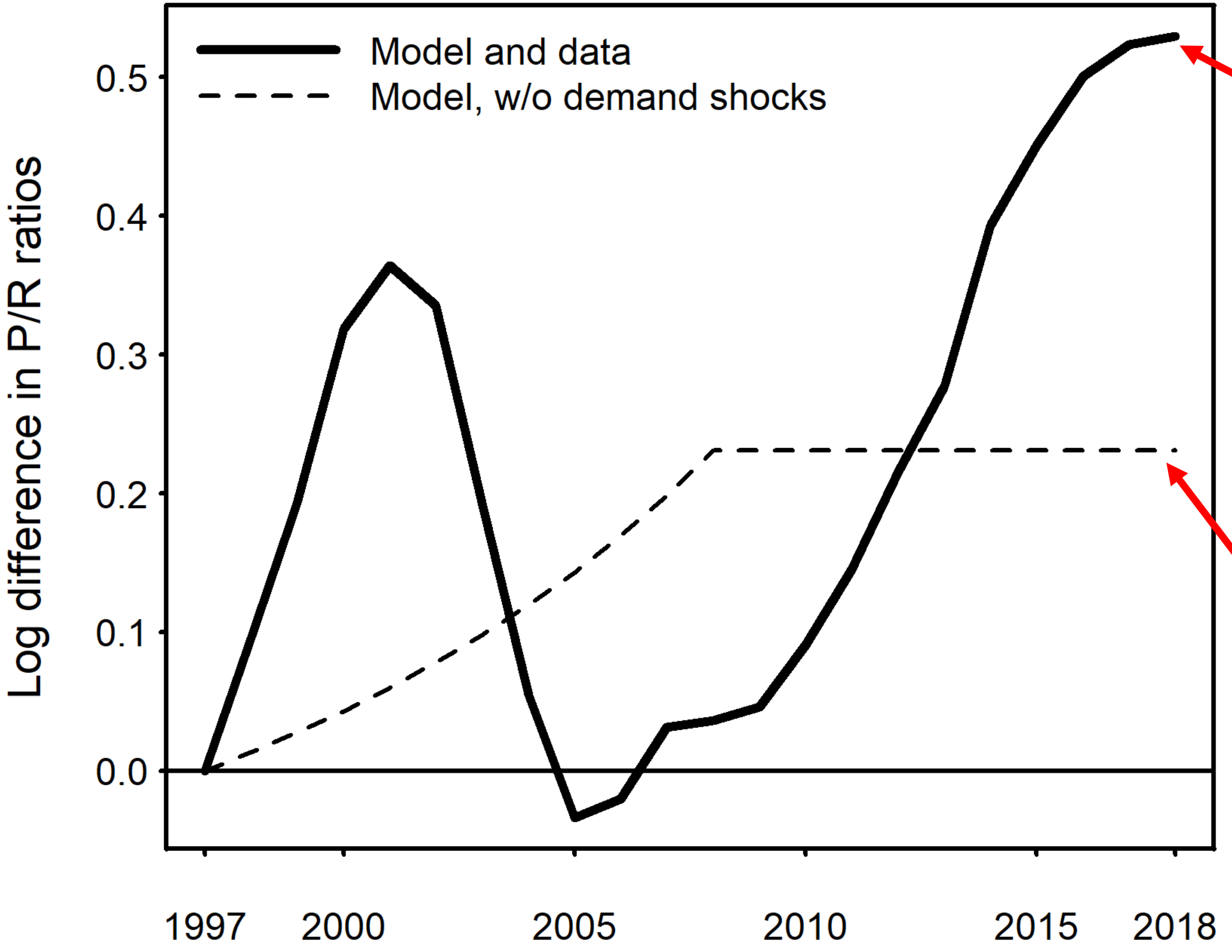




Grey shading: All, one or no LPA experienced declining demand

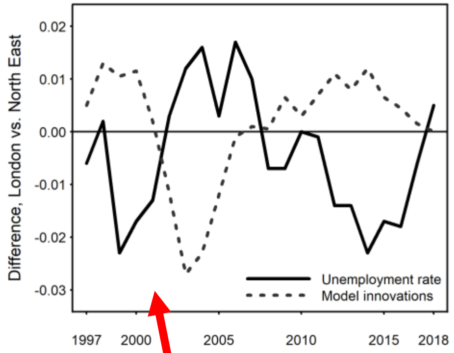
Difference in log P-R-ratio b/w London and North East

(Based on model simulation w. infinite no. of periods &
model parameters inferred from data and empirical analysis)



Actual difference in log P-R-ratio b/w two regions can be fully explained by model with spatially varying demand shocks, spatial variation in supply elasticity & autocorrelation in demand shocks

What changing discount rates alone can explain



Model implied difference in demand shocks nearly perfectly correlated with observed difference in unemployment rates

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Commercial vs. residential real estate cycles

- **Residential RE cycles**
  - Often related to altering economic demand shocks

- **Office and Retail**
  - Cycles often bear almost no relation to broader economic cyclicality
  - Cycles are often very pronounced

⇨ **Why? (Discuss)**

Differences in involved agents & assets…

- For **commercial investors** (relative to home-buyers)
  - Investment motive more important
  - More myopic?
  - Liquidity constraints & loss aversion less relevant

Differences in involved agents & assets…

- Office buildings tend to have longer time lags
  - Longer planning & construction lags encourage endogenous oscillations
- Office space subject to much longer leases
  - Encourages endogenous oscillations
- Retail space typically has short durability
  - Encourages endogenous oscillations (see Wheaton 1999)
- Office and retail markets are subject to greater demand volatility
  - Increases value of option to wait & profitability of excess capacity

Conclusions & policy implications

1. **Housing cycles** can often be explained by altering economic demand shocks (business cycles) in conjunction with **inelastic long-run supply**

2. **Policy implication**: In places with inelastic supply – be cautious with place-based policies ⇨ general rule (with caveats): ‘help people—not places’

3. Some cycles – especially commercial ones – are ‘**endogenously driven**’

Conclusions & implications *(cont.)*

4. **Office & retail cycles** often bear almost no relation to broader economic cyclicality
   - Cycles triggered by initial economic shock
   - Causes oscillations to eventually revert to long-run trend
   - Cycles often very pronounced
5. Inelastic supply in conjunction with autocorrelated demand shocks can, to good extent, explain **why prices have risen so much more than rents** over past 20 years in superstar cities
6. Residential and commercial RE cyclicality **differs because involved agents and assets differ**

⇨ **No single theory can explain all phenomena; many factors drive real estate cycles!**

Preparations for *Block III*—**Seminar 6** (Week 10)

- Topic: Should governments subsidize homeownership?
- Seminar format: Key paper presentations (see handout on Moodle)
- Papers to be discussed: DiPasquale and Glaeser (1999) and Hilber and Turner (2014)
- Tasks for all students: Read key papers and prepare questions for presenters